Stochastic volatility latent states for IBM options

Stochastic Volatility Latent States for IBM Options: Key Methods and Findings

Research on stochastic volatility with latent states focuses on how to infer unobserved volatility from prices and use it for option valuation and risk analysis. Several papers provide tools and insights relevant to IBM options and more general equity/index options.

Latent Volatility for IBM Options

A high-frequency instrumental-variables framework links many realized volatility (RV) measures and option-implied volatilities to a **latent volatility process** for IBM from 2009–2013 (Ahsan & Dufour, 2025). Identification-robust tests show this latent volatility is **highly persistent, close to unit-root**, with the persistence parameter around 0.9–1.0 (Ahsan & Dufour, 2025). Five-minute RV measures and IBM call option implied volatilities are the most informative instruments; too-high-frequency RVs inflate confidence interval width due to microstructure noise (Ahsan & Dufour, 2025).

Latent State Inference Approaches

Approach / Model feature	How latent volatility/state is extracted	Citations
HF IV methods on IBM	Instrumental variables using RV + IVs to infer low-frequency latent volatility and its persistence	(Ahsan & Dufour, 2025)
Realized measures + returns	Dual measurement equations and filtering/smoothing for latent conditional variance	(Bormetti et al., 2019; Corsi et al., 2013)
Markov switching SV	EM/Baum–Welch + VIX to filter hidden Markov chain and invert latent volatility	(Goutte et al., 2017)
Particle / Bayesian filters	EKF/UKF/PF and adaptive switching for volatility and “risk” states from option prices; particle filters for long-memory SV	(Yashaswi, 2021; Chronopoulou & Viens, 2012)
Closed-form filtering for continuous-time SV	Transition-density-based filters and likelihood, without proxies or simulation	(Aït-Sahalia et al., 2020)

FIGURE 1 Overview of methods to infer latent volatility states from prices and options

Option Pricing with Latent Stochastic Volatility

Several models explicitly incorporate unobserved volatility into option pricing:

- Discrete-time SV models with realized measures allow **semi-analytical pricing** and show that filtering/smoothing RV measures increases estimated volatility persistence, crucial for matching S&P 500 option prices (Bormetti et al., 2019).
- A long-memory SV model and its discrete approximations use particle filtering to estimate the latent volatility distribution, then build multinomial trees for option pricing on S&P 500 (Chronopoulou & Viens, 2012).
- Models using realized volatility as a proxy for latent volatility, with long-memory dynamics and analytic change of measure, outperform competing SV models in pricing S&P 500 options (Corsi et al., 2013).
- Regime-switching SV with CIR variance and hidden Markov chain uses maximum likelihood and Fourier methods to price and calibrate S&P 500 and VIX options, fitting joint stock/volatility features well (Goutte et al., 2017).

Estimation and Filtering of Latent States

- Approximate Bayesian computation and particle MCMC enable filtering and parameter estimation for **α-stable SV** models with intractable likelihoods (Vankov et al., 2019).
- An efficient quasi-Bayesian LTE/SMC algorithm uses **risk-neutral cumulants to marginalize latent states** in affine option models with multiple volatility factors and jumps, performing well on large option panels (Brignone et al., 2022).
- Closed-form approximations provide maximum likelihood estimators and **filters for latent states** in continuous-time SV and term-structure models, with linear computational cost and proven convergence as observation frequency increases (Aït-Sahalia et al., 2020).

Summary

Work directly on IBM finds its latent volatility, inferred from high-frequency measures and IBM options, is extremely persistent, near a unit-root process. More broadly, modern stochastic volatility option models treat volatility as a latent state inferred via instruments, realized measures, particle/Bayesian filters, EM algorithms, or closed-form filter recursions. These latent-state methods generally improve option pricing accuracy and the match to implied volatility surfaces compared with constant-volatility models.

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